

ADVANCED POINTERS

```
1.#include<stdio.h>
void main()
{
    int i=10;
    int *p=i;
    char **q=&p;
    printf("%d\n",*q+1);
    printf("%d\n",**q);
}
```

```
2.#include<stdio.h>
void main()
{
    int i=300, j=600;
    char *p[2]={&i,&j};
    short int **q=p+1;
    p[1][1]=0;
    printf("%d",**q);
}
```

```
3.#include<stdio.h>
void main()
{
    char **q=1000;
    printf("%d",q+1);
    printf("%d",**q);
}
```

```
4.#include<stdio.h>
void main()
{
    int a[5]={ 10,20,30,40};
    char *p[3]={a+2,a+1,a};
    short int **q=p+1;
    ++*q;
    printf("%d\n",q[1][1]);
    printf("%d\n",q[0][-1]);
}
```

```
5.#include<stdio.h>
void main()
{
    int i=300,j=400;
    char *p=&j;
    short int **q=&p;
    *q=&i;
    ++*q;
    *--*q=*(p+1);
}
```

```
    printf("%d\n",**q);
}
```

```
6.#include<stdio.h>
void main()
{
    int a[5]={ 10,20,30,40,50};
    char *p=&a[1];
    short int *q=&a[4];
    short int **s=&p;
    short int **b=&q;
    printf("%d\n",*b-*s);
}
```

```
7.#include<stdio.h>
void main()
{
    int i=300, j=322;
    short int *p=&i;
    char **q=&p;
    short int ***r= &q;
    ++**r;
    printf("%d\n",p[-1]);
    printf("%d\n",p[0]);
}
```

```
8.#include<stdio.h>
void main()
{
    int a[2][5]={ {65,66,67},{49,50,51,52}};
    short int (*p)[1]=a;
    p[1][-1]=(a[1][0]<<3)|(a[1][1]);
    printf("%d",**p);
}
```

```
9.#include<stdio.h>
void main()
{
    int a[2][5]={ {65,66,67},{49,50,51,52}};
    short int (*p)[1]=a;
    printf("%d %d",p[1][0],p[5][-1]);
}
```

```
10.#include<stdio.h>
void main()
{
    int a[2][5]={ {65,66,67},{49,50,51,52}};
    short int *q=a;
    printf("%d\n",*q);
    printf("%d\n",*(q+1));
}
```

```
}
```

```
11.#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int a[6]={};
```

```
    short int *p[3]={a+2,a+1,a};
```

```
    p[2][1]=65|66<<8;
```

```
    char *q=a+1;
```

```
    short int **r=&q;
```

```
    r[0][-2]=49|50<<8;
```

```
    printf("%s",a);
```

```
}
```

```
12.#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int a[5]='A','B','C','D','E';
```

```
    char (*p)[2]=a+1;
```

```
    short int *q=a+1;
```

```
    *q=50;
```

```
    p[0][1]=51;
```

```
    p[1][0]=52;
```

```
    p[1][1]=53;
```

```
    printf("%s",a+1);
```

```
}
```

```
13.#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int i=0;
```

```
    char a[3][10]={"ABCDE","PQRSTUV","123456"};
```

```
    short int (*p)[3]=a;
```

```
    p[2][2]=p[4][0];
```

```
    printf("%s",a[1]);
```

```
}
```

```
14.#include<stdio.h>
```

```
void main()
```

```
{
```

```
    int a[6]={ 10,20,30,40,50,60};
```

```
    short int (*p)[2]=a;
```

```
    p[1][-1]=258;
```

```
    printf("%d %d",p[0][0],p[0][1]);
```

```
}
```

```
15.#include<stdio.h>
```

```
void main()
```

```
{
```

```
    char a[6]={ 10,20,30,40,50,60};
```

```

    short int (*p)[2]=a;
    p[1][-1]=258;
    printf("%d %d",a[2],a[3]);
}

```

```

16.#include<stdio.h>
void main()
{
    short int *p[3]={ "ABCDEF","PQRSTUV","123456"};
    char **q=p+1;
    p[1][2]**q;
    printf("%s",p[1]);
}

```

```

17.#include<stdio.h>
void main()
{
    char p[3][8]={ "ABCDEF","PQRSTUV","123456"};
    char *q=p+1;
    p[1][2]*q;
    printf("%s",p[1]);
}

```

```

18.#include<stdio.h>
void main()
{
    char a[10]='1','2','3','4','5';
    int (*p)[3]=a;
    printf("%d",*(p+5)-*p);
}

```

```

19.#include<stdio.h>
void main()
{
    short int *a[3]={ 1000,2000,4000};
    printf("%d",*(a+2)-*a);
}

```

```

20.#include<stdio.h>
void main()
{
    int i=500,j=450,k=258;
    short int *a[3]={&i,&j,&k};
    char *p;
    p=++*(a+2);
    p[-1]=2;
    printf("%d",k);
}

```

```

21.#include<stdio.h>

```

```

void main()
{
    int a[5]={1,2,3,4,5};
    short int (*p)[4]=a;
    printf("%p\n",a);
    printf("%p\n",p+4);
    printf("%p",*p+4);
}

```

```

22.#include<stdio.h>
void main()
{
    short int *p[3]={"ABCDE","PQRST","12345"};
    p[1]="6789T";
    printf("%s",p[1]+1);
}

```

```

23.#include<stdio.h>
void main()
{
    double a[5]={10,20,30,40,50};
    char *p=a+1;
    short int *q=p+20;
    int **m=&p;
    int **n=&q;
    printf("%d",*n-*m);
}

```

```

24.#include<stdio.h>
void main()
{
    int i=400;
    char *p=&i;
    short int **q=&p;
    *++*q=0;
    printf("%d\n",**q);
    printf("%d\n",q[0][-1]);
    q[0][-1]**q;
    printf("%d",**q);
}

```

```

25.#include<stdio.h>
void main()
{
    short int a[10]={0,2,0,4,5};
    char *p=a+1;
    int **q=&p;
    printf("%d\n",q[0][2]);
    --*q;
    printf("%d\n",q[0][1]);
}

```

```
}
```

```
26.#include<stdio.h>
void main()
{
    char a[10]="ABCDEFGH";
    short int (*p)[2]=a;
    p[2][-2]=49;
    printf("%s\n",a);
}
```

```
27.#include<stdio.h>
void main()
{
    int a[5]={10,20,30,40,50};
    char (*p)[3]=a;
    p[3][-4]=1;
    printf("%d\n",a[1]);
}
```

```
28.#include<stdio.h>
void main()
{
    int i=350,j=450,k=550;
    short int * p[3]={&i,&j,&k};
    ++*(p+1);
    printf("%d\n",p[1][-1]);
    printf("%d\n",p[1][0]);
}
```

```
29.#include<stdio.h>
void main()
{
    int i=10;
    int *p=i;
    printf("%d\n",(char *)p+1);
    printf("%d\n",p+1);
}
```

```
30.#include<stdio.h>
void main()
{
    int i=20;
    short int *p=i;
    int *q=i;
    char *r=i;
    printf("%d\n",p+1);
    printf("%d\n",q+1);
    printf("%d\n",r+1);
}
```

```

31.#include<stdio.h>
void main()
{
    int i=400;
    char *p=&i;
    ++p;
    *p=0;
    printf("%d\n",i);
}

```

```

32.#include<stdio.h>
void main()
{
    int i=500;
    short int *p=&i;
    ++p;
    *(p-1)=256;
    printf("%d %d\n",*p,i);
}

```

```

33.#include<stdio.h>
void main()
{
    int i=600;
    short int *p=&i;
    char *q=&i;
    q=q+2;
    *--q=1;
    printf("%d \n",*p);
}

```

```

34.#include<stdio.h>
void main()
{
    char ch='F';
    int *p=ch;
    printf("%u \n",p);
    printf("%u \n",&ch);
    printf("%u \n",&ch+'F');
    printf("%u \n",p+'F');
}

```

```

35.#include<stdio.h>
void main()
{
    double a[5]={10,20,30,40,50};
    char *p=a+1;
    short int *q=a+4;
    printf("%lf\n",a[1]);
    printf("%d\n",q-(short int *)p);
}

```

```

        *p=q-(short int *)p;
        printf("%lf",a[1]);
    }

```

```

36.#include<stdio.h>
void main()
{
    int a[5]={10,20,30,40,50};
    short int *p=a+1;
    char *q=a;
    ++p;
    p[-3]=33;
    printf("%d",*q);
}

```

```

37.#include<stdio.h>
void main()
{
    short int a[5]={1,2,3,4,5};
    int *p=a;
    *p=258;
    printf("%d %d\n",a[0],a[1]);
}

```

```

38.#include<stdio.h>
void main()
{
    int a[5]={1,2,3,4,5};
    char *p=a+1;
    ++p;
    printf("%d \n",(*(&p))[5]);
}

```

```

39.#include<stdio.h>
void main()
{
    int a[5]={48,57,56};
    int *p=a;
    *p|=49<<8|50<<16;
    printf("%s \n",a);
}

```

```

40.#include<stdio.h>
void main()
{
    char a[8]="ABCD";
    short int *p=a;
    p[2]=49|50<<8;
    printf("%s \n",a);
}

```



```

41.#include<stdio.h>
void main()
{
    char a[12]="ABCDCE12345";
    int *p=a+8;
    p[-1]=0;
    char *q=a;
    printf("%s %s\n",p,q+2);
}

```

```

42.#include<stdio.h>
void main()
{
    extern double *p;
    printf("%d\n",p+30);
}
double *p;

```

```

43.#include<stdio.h>
int *p;
void main()
{
    static int i;
    if(i<5)
    {
        ++p;
        ++i;
        main();
        printf("%d\n",p);
    }
}

```

```

44.#include<stdio.h>
void main()
{
    int i=260;
    short int *p=&i;
    short int *q=&i+1;
    printf("%d \n",q-p);
}

```

```

45.#include<stdio.h>
void main()
{
    int i=10;
    char *p=i;
    short int **q=&p;
    printf("%d\n",p+1);
    printf("%d\n",q+1);
    printf("%d\n",*q+1);
}

```

```
}
```

```
46.#include<stdio.h>
void main()
{
    char s[10]="ABCDEPQ";
    int *p=s;
    short int **q=&p;
    q[0][2]=50|49<<8;
    printf("%s\n",s);
}
```

```
47.#include<stdio.h>
void main()
{
    int i=400;
    short int *p=&i;
    char **q=&p;
    *++*q=2;
    q[0][-1]=0;
    printf("%d\n",i);
}
```

```
48.#include<stdio.h>
void main()
{
    int i=300,j=400,k=500;
    short int * p[3]={&i,&j,&k};
    char **q[3]={p+2,p+1,p};
    *(q+1)=p+2;
    q[1][-1][1]=0;
    printf("%d\n",j);
    printf("%d\n",q[1][-1][0]);
}
```

```
49.#include<stdio.h>
void main()
{
    int i=0;
    char a[3][10]={"ABCDE","PQRSTUV","123456"};
    char *p[3]={a[0],a[1],a[2]};
    char **q[3]={p,p+2,p+1};
    for(i=1;a[1][i];i=i+2)
        q[1][-1][i]=(48+i);
    printf("%s",a[1]);
}
```

```
50.#include<stdio.h>
void main()
{
```

```
int a[2][2]={1,2,3,4};
int *p[2]={a[1],a[1]+1};
int **q[2]={p+1,p};
*(q+1)=p+1;
printf("%d\n",*q[1][-1]);
}
```